Operating Instructions

Diesel engine

16V2000G56S

MS150117/04E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
16V2000G56S	50.5 kW/cyl.	3B, continuous service, variable load, ICXN

Table 1: Applicability

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

⚠ **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

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2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:http://www.mtu-solutions.com

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2.2 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:

- The installation of the engine
- The design of the engine-generator set
- The installation of the engine-generator set

The relevant standards are, in particular:

- DIN EN 1679-1
- EN ISO 8528-13

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. We therefore recommend a redundant design of the engine-generator set system in safety-critical applications.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, exhaust turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).

- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:

- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary and suitable safety devices to reliably exclude the possibility of personal injury, e.g.:

- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these protective devices. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the protective devices.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Chemical resistance

Only use hoses, pipes, fittings, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

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Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.

Incorrect synchronization

Load application with incorrect synchronization (generator connection with phase offset) can cause considerably high torque pulses. The plant integrator must design the control system such that the possibility of a load application with incorrect synchronization is excluded.

Emergency stop

The diesel engine has inputs for Stop and Emergency stop. The emergency-stop function must be executable in the whole plant independently of the operation stop. This is the only way to ensure that a safe shutdown of fuel injection takes place on a diverse redundant path and thus results in engine standstill. Through an appropriate design of the engine-generator set or whole plant, the operator of the engine must guarantee that safe, non-reversible stopping of the engine is as fast as possible, including all of its dangerous system components, when the emergency-stop actuating element is activated. In particular, it must always be guaranteed that machine shutdown can not be prevented or delayed once the command for shutdown has been issued. The engine shutdown command must always have priority over the start command.

If machines or machine components are designed to interact, they must be designed and built such that all devices can shut down. This relates to:

- Emergency-stop command devices for the machine itself (engine, engine-generator set)
- All connected devices insofar as their continued operation could result in a hazard

Important
Engines with exhaust aftertreatment: When the emergency stop is actuated on engine with an exhaust aftertreatment system, note that the emergency-stop function also interrupts the reducing agent supply. This can result in overheating of the reducing agent dosing units. For details and further information, see Operating Instructions.

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